

Four Motzkin Congruences

A Constant-Term Proof with Sharp Ranges

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Abstract

The OEIS entry for the Motzkin numbers records four congruences proposed by Peter Bala on 10 February 2022. We give self-contained proofs of all four. Two concern indices immediately preceding multiples of prime powers; two assert repetition of an initial block modulo a prime or its square. A single constant-term transfer theorem explains the block phenomena at arbitrary prime-power precision. For the prime-square case, the arithmetic input is obtained by factoring a quadratic over the sixth roots of unity and proving a binomial coefficient congruence directly. We further show that the two proposed initial-block ranges are optimal for every prime and exponent in their respective statements, and compute the first nonzero error explicitly. The boundary-index formulas are extended to all powers of every odd prime. A digit algorithm evaluates a Motzkin number modulo an odd prime without constructing the integer itself. Reproducible Python programs accompany the article; 276,814 finite checks pass. The proofs are mathematical arguments, not deductions from these computations. Related results of Kohen and Pan–Sun are identified explicitly; no claim of publication priority is inferred from an OEIS conjecture label.

Main outcome. All four targeted formulas are true. In particular, for every prime $p \geq 5$ and $k \geq 2$,

$$M_{p^k+n} \equiv M_n \pmod{p^2} \quad (0 \leq n \leq p^{k-1} - 3),$$

whereas at the very next index the difference is

$$M_{p^k+p^{k-1}-2} - M_{p^{k-1}-2} \equiv \begin{cases} -2p, & p \equiv 1 \pmod{6}, \\ p, & p \equiv 5 \pmod{6} \end{cases} \pmod{p^2}.$$

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1 The sequence and the four targets

1.1 Why the Motzkin numbers are a substantial target

The Motzkin number M_n counts paths from $(0, 0)$ to $(n, 0)$ whose steps are $(1, 1)$, $(1, 0)$, and $(1, -1)$, and whose height never becomes negative. The sequence starts

$$1, 1, 2, 4, 9, 21, 51, 127, 323, 835, 2188, 5798, \dots$$

It is A001006. Its many interpretations include noncrossing partial matchings and restricted trees and permutations; the entry provides references and numerous equivalent formulas [1]. Thus the target is a classical combinatorial sequence, not an isolated artificial example.

Our auxiliary sequences are the central and adjacent trinomial coefficients. Put

$$P(x) = x^{-1} + 1 + x, \quad F(x) = 1 + x + x^2 = xP(x), \quad Q(x) = 1 - x^2, \quad (1.1)$$

and define

$$T_n = \text{CT } P(x)^n, \quad U_n = [x^1]P(x)^n \quad (n \geq 0). \quad (1.2)$$

Here CT extracts the coefficient of x^0 from a Laurent polynomial. The T_n form A002426, and the U_n for $n \geq 1$ form A005717; we explicitly set $U_0 = 0$ [2, 3]. Also write

$$D_0 = 1, \quad D_n = T_{n-1} + U_{n-1} \quad (n \geq 1). \quad (1.3)$$

This is A005773, the directed-animal counting sequence; the identity in (1.3) is recorded in that entry [4]. The present proof needs only its coefficient formulation.

1.2 The exact statements to be proved

The following are the four formulas in Bala's comment dated 10 February 2022 in A001006, with notation replaced by that just introduced [1]. The quantifier on n is important, as are the upper endpoints of the two ranges.

(B1) Boundary indices, first prime class.

If $p \equiv 1 \pmod{6}$ is prime and $n, r \geq 1$, then

$$M_{np^r-2} \equiv -U_{n-1} \pmod{p}. \quad (\text{B1})$$

(B2) Boundary indices, second prime class.

If $p \equiv 5 \pmod{6}$ is prime and $n \geq 1$, then

$$M_{np-2} \equiv -D_n \pmod{p}. \quad (\text{B2})$$

(B3) Repetition modulo a prime.

If $p \geq 3$ is prime and $k \geq 1$, then

$$M_{n+p^k} \equiv M_n \pmod{p} \quad (0 \leq n \leq p^k - 3). \quad (\text{B3})$$

(B4) Repetition modulo the square of a prime.

If $p \geq 5$ is prime and $k \geq 2$, then

$$M_{n+p^k} \equiv M_n \pmod{p^2} \quad (0 \leq n \leq p^{k-1} - 3). \quad (\text{B4})$$

The proofs below resolve precisely these four assertions, not every conjecture mentioned elsewhere in the Motzkin entry.

1.3 Status, attribution, and what is established here

The four formulas above were still grouped under a conjecture heading in the OEIS entry consulted for this article. A database label is not a reliable certificate that a statement is absent from the literature. The entry itself includes a November 2024 comment by Rob Burns pointing out that the $n = 1$ boundary cases were already consequences of his 2017 work [1]. Kohen's 2024 work develops constant-term and base- p methods for Motzkin and central trinomial numbers [5, 6]. In particular, the prime-modulus formulas here overlap with consequences of his digit decompositions and boundary identities. Pan and Sun establish considerably stronger central-trinomial congruences than the prime-square input needed here [7]; their paper first appeared as a preprint in 2020 and its final version appeared in 2026.

The contribution of this note is a complete, directly checkable treatment of the four OEIS-listed formulas together, including the extension to higher prime powers, the optimality of both initial-block ranges, and the explicit first errors. Every lemma needed for (B1)–(B4) is proved below. Neither the proofs nor the claims of sharpness depend on a literature search establishing novelty. No priority claim is made for the auxiliary congruences or for consequences already implicit in the cited work.

2 Constant terms and elementary identities

2.1 A reflection argument

Lemma 2.1. *For all $n \geq 0$,*

$$M_n = \text{CT } Q(x)P(x)^n = [x^n](1 - x^2)F(x)^n. \quad (2.1)$$

Proof. The coefficient of x^h in $P(x)^n$ counts unrestricted paths of length n ending at height h . Thus T_n counts all paths ending at height zero. A path of this kind that reaches height -1 has a first visit to -1 . Exchange its up-steps and down-steps through that visit and leave all later steps unchanged. The transformed path ends at height 2. Conversely, any unrestricted path ending at height 2 has a first visit to height 1; exchanging up and down through that visit recovers the bad path. Horizontal steps are unchanged in both directions.

Consequently $M_n = [x^0]P^n - [x^2]P^n$. The polynomial P^n is symmetric under $x \mapsto x^{-1}$, so this difference is $\text{CT}(1 - x^2)P^n$. Multiplication by x^n gives the second expression. $\square$

For later use observe the exact support bound

$$\text{supp}(Q(x)P(x)^n) \subseteq [-n, n + 2] \cap \mathbb{Z}. \quad (2.2)$$

It is the extra two units at the right edge, caused by $-x^2$, that produce both the range cutoffs and their first corrections.

2.2 The three neighboring coefficients

Multiplication by $P(x)$ and symmetry give

$$T_{n+1} = T_n + 2U_n. \quad (2.3)$$

Writing $V_n = [x^2]P(x)^n$, we also have $T_{n+2} = 3T_n + 4U_n + 2V_n$. Since $M_n = T_n - V_n$, elimination yields

$$\boxed{2M_n = 3T_n + 2T_{n+1} - T_{n+2}}. \quad (2.4)$$

This is the identity that will turn the digit rule for T_n into a fast algorithm for M_n .

2.3 Generating functions and exact recurrences

Selecting the up- and down-steps of an unrestricted path gives

$$T_n = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \binom{2j}{j}. \quad (2.5)$$

Consequently, as formal power series,

$$\begin{aligned} \mathcal{T}(z) &= \sum_{n \geq 0} T_n z^n = \frac{1}{1-z} \sum_{j \geq 0} \binom{2j}{j} \left(\frac{z^2}{(1-z)^2} \right)^j \\ &= \frac{1}{\sqrt{1-2z-3z^2}}. \end{aligned} \quad (2.6)$$

Differentiation gives

$$(n+1)T_{n+1} = (2n+1)T_n + 3nT_{n-1} \quad (n \geq 1). \quad (2.7)$$

A nonempty Motzkin path either begins with a horizontal step, or has a unique decomposition $U\pi D\rho$, where D is the step of its first return to height zero. Both π and ρ are Motzkin paths after a vertical translation when needed. Therefore

$$\mathcal{M}(z) = 1 + z\mathcal{M}(z) + z^2\mathcal{M}(z)^2 = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}. \quad (2.8)$$

Deleting horizontal steps from a path with j up-steps leaves a Dyck path of semilength j . Hence

$$M_n = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \frac{1}{j+1} \binom{2j}{j}. \quad (2.9)$$

From (2.8), differentiation and simplification give

$$z(1 - 2z - 3z^2)\mathcal{M}'(z) + (2 - 3z - 3z^2)\mathcal{M}(z) = 2.$$

Comparing coefficients yields the exact integer recurrence

$$(n+2)M_n = (2n+1)M_{n-1} + 3(n-1)M_{n-2} \quad (n \geq 2), \quad M_0 = M_1 = 1. \quad (2.10)$$

These recurrences are useful for exact verification. In modular arithmetic their denominators must not be inverted when divisible by the modulus's prime; this point matters in the supplied implementation.

3 Prime powers spread the exponents apart

3.1 Lifting a polynomial congruence

All congruences between Laurent polynomials in this section are coefficientwise. They take place in $\mathbb{Z}[x, x^{-1}]$.

Lemma 3.1. *Let p be prime. If $A \equiv B \pmod{p^a}$ for $a \geq 1$, then $A^p \equiv B^p \pmod{p^{a+1}}$.*

Proof. Write $A = B + p^a C$ and expand. The $j = 1$ term is divisible by p^{a+1} . For $1 < j < p$, the factor $\binom{p}{j}$ contributes p , and the other factor contributes p^{aj} , which is more than sufficient. The final term is divisible by p^{ap} , and $ap \geq a + 1$. $\square$

Lemma 3.2. *For a Laurent polynomial L with integer coefficients, a prime p , and integers $1 \leq s \leq k$, put $q = p^{k-s+1}$. Then*

$$L(x)^{p^k} \equiv L(x^q)^{p^{s-1}} \pmod{p^s}. \quad (3.1)$$

Proof. The Frobenius congruence in characteristic p gives $L(x)^q \equiv L(x^q) \pmod{p}$. Applying theorem 3.1 successively $s - 1$ times raises the exponent on both sides to p^{s-1} and the modulus to p^s . $\square$

Notice the difference between precision and shift. To retain s digits of p -adic precision, we replace the large exponent p^k by the smaller exponent p^{s-1} , but the surviving exponents of x are then spaced by $q = p^{k-s+1}$. Only the coefficients near exponent zero can interact with a sufficiently short factor QP^n .

3.2 The full local-block theorem

Theorem 3.3 (Prime-power block transfer). *Let p be an odd prime, $m \geq 0$, and $1 \leq s \leq k$. Set*

$$q = p^{k-s+1}, \quad h = mp^{s-1}.$$

Then, for $0 \leq d < q$,

$$M_{mp^k+d} \equiv \begin{cases} T_h M_d, & 0 \leq d \leq q-3, \\ T_h M_{q-2} - U_h, & d = q-2, \\ T_h M_{q-1} - (q-1)U_h, & d = q-1 \end{cases} \pmod{p^s}. \quad (3.2)$$

Proof. By theorem 3.2, raising to the nonnegative integer power m gives

$$P(x)^{mp^k} \equiv P(x^q)^h \pmod{p^s}.$$

Thus theorem 2.1 gives

$$M_{mp^k+d} \equiv \text{CT}(Q(x)P(x)^d P(x^q)^h) \pmod{p^s}. \quad (3.3)$$

Every exponent in the last factor is a multiple of q .

If $d \leq q - 3$, the support of QP^d lies strictly between $-q$ and q , by (2.2). Only its constant term can meet an opposite exponent from the last factor. The constant term of $P(x^q)^h$ is T_h , and the first case follows.

If $d = q - 2$, the support is contained in $[-q + 2, q]$. Besides the constant term, exactly one possible multiple of q remains, namely q . Its coefficient is -1 : it comes from $-x^2$ times the top monomial x^{q-2} of P^{q-2} . The coefficient of x^{-q} in $P(x^q)^h$ is U_h by symmetry. This gives the second case.

If $d = q - 1$, the support is contained in $[-q + 1, q + 1]$. Since $q \geq 3$, again the only nonzero multiple of q that can occur is q . Its coefficient is

$$[x^q]QP^{q-1} = -[x^{q-2}]P^{q-1} = -(q-1).$$

Indeed, an exponent one below the maximum is obtained by choosing the constant monomial from exactly one of the $q - 1$ factors, and x from all the others. Substitution in (3.3) gives the last case. $\square$

Remark 3.4. The theorem is uniform in m and contains all three residues near the end of a block. It is not an assertion of global periodicity. In the interior, the multiplying factor is a central trinomial coefficient; at the boundary, an adjacent coefficient enters as well.

3.3 Proof and exact range of (B3)

Taking $s = 1$ and $m = 1$ in theorem 3.3 gives $q = p^k$, $h = 1$, and $T_1 = U_1 = 1$. The interior case is exactly (B3). The next case gives

Corollary 3.5. *For every odd prime p and $k \geq 1$,*

$$M_{2p^k-2} - M_{p^k-2} \equiv -1 \pmod{p}. \quad (3.4)$$

Consequently $p^k - 3$ is the largest upper endpoint L for which $M_{p^k+n} \equiv M_n \pmod{p}$ holds for every $0 \leq n \leq L$.

For example, when $p = 5$ and $k = 1$, the differences for $n = 0, 1, 2$ vanish modulo 5, whereas $M_8 - M_3 = 323 - 4 = 319 \equiv -1 \pmod{5}$.

4 Boundary residues and proofs of (B1) and (B2)

4.1 An elementary period-three coefficient sequence

For an odd prime p , define

$$\varepsilon_p = \begin{cases} 0, & p = 3, \\ 1, & p \equiv 1 \pmod{3}, \\ -1, & p \equiv 2 \pmod{3}. \end{cases} \quad (4.1)$$

No quadratic reciprocity is needed for the following calculation. We use only

$$\frac{1}{F(x)} = \frac{1-x}{1-x^3} = \sum_{j \geq 0} c_j x^j, \quad c_j = \begin{cases} 1, & j \equiv 0 \pmod{3}, \\ -1, & j \equiv 1 \pmod{3}, \\ 0, & j \equiv 2 \pmod{3}. \end{cases} \quad (4.2)$$

Lemma 4.1. *For every odd prime p , every $r \geq 1$, and $q = p^r$,*

$$T_{q-1} \equiv \varepsilon_p^r \pmod{p}, \quad (4.3)$$

$$M_{q-2} \equiv \frac{\varepsilon_p^r - 1}{2} \pmod{p}, \quad (4.4)$$

$$M_{q-1} \equiv \frac{1 + 3\varepsilon_p^r}{2} \pmod{p}. \quad (4.5)$$

The fractions mean multiplication by the inverse of 2 modulo p .

Proof. Since $F(x)^q \equiv F(x^q) \pmod{p}$ and $F(0) = 1$, division by F or F^2 is legitimate in $\mathbb{F}_p[[x]]$. At any degree below q , the factor $F(x^q) = 1 + x^q + x^{2q}$ contributes only its constant term. Therefore

$$T_{q-1} = [x^{q-1}]F^{q-1} \equiv [x^{q-1}]F^{-1} = c_{q-1}.$$

If $p \neq 3$, the residue of $q = p^r$ modulo 3 is determined by the parity of r when $p \equiv 2 \pmod{3}$; the result is ε_p^r . If $p = 3$, the coefficient is $c_{q-1} = 0$. This proves (4.3).

For the second identity, (2.1) gives

$$M_{q-2} \equiv [x^{q-2}] \frac{1-x^2}{F(x)^2}.$$

The rational function here is a derivative:

$$\frac{1-x^2}{F(x)^2} = \left(\frac{x}{F(x)} \right)'. \quad (4.6)$$

Thus $M_{q-2} \equiv (q-1)c_{q-2} \equiv -c_{q-2} \pmod{p}$. If $q \equiv 1 \pmod{3}$ this is zero; if $q \equiv 2 \pmod{3}$ it is -1 . For $p = 3$ it is 1, which equals $-1/2 \pmod{3}$. These are precisely (4.4).

Finally,

$$M_{q-1} \equiv [x^{q-1}] \frac{1-x^2}{F(x)} = c_{q-1} - c_{q-3}.$$

The three cases are respectively 2, -1 , and -1 when q is 1, 2, or 0 modulo 3. They give (4.5), including $p = 3$ where $1/2 \equiv -1 \pmod{3}$. $\square$

4.2 A single formula covering every odd prime

Theorem 4.2 (Boundary-index formula). *For every odd prime p and every $n, r \geq 1$,*

$$M_{np^r-2} \equiv \frac{\varepsilon_p^r - 1}{2} T_{n-1} - U_{n-1} \pmod{p}. \quad (4.7)$$

Proof. Use theorem 3.3 with precision $s = 1$, exponent $k = r$, multiplier $m = n - 1$, and offset $d = p^r - 2$. Its second case gives

$$M_{np^r-2} \equiv T_{n-1}M_{p^r-2} - U_{n-1} \pmod{p}.$$

Now use (4.4). $\square$

If $p \equiv 1 \pmod{6}$, then $\varepsilon_p = 1$ and the first term vanishes. This proves (B1), including $n = 1$ because $U_0 = 0$. If $p \equiv 5 \pmod{6}$ and $r = 1$, then $\varepsilon_p = -1$, so the right-hand side is $-T_{n-1} - U_{n-1} = -D_n$. This proves (B2).

The same theorem provides a useful extension that is not present in the wording of (B2):

$$M_{np^r-2} \equiv \begin{cases} -D_n, & p \equiv 5 \pmod{6}, r \text{ odd}, \\ -U_{n-1}, & p \equiv 5 \pmod{6}, r \text{ even} \end{cases} \pmod{p}. \quad (4.8)$$

At $p = 3$ it becomes

$$M_{n3^r-2} \equiv T_{n-1} - U_{n-1} \pmod{3}. \quad (4.9)$$

Thus simply replacing p by p^r on the left of (B2), without regard to the parity of r , would give the wrong result.

5 The prime-square arithmetic input

The transfer theorem at precision two leaves the scalar T_{mp} . We now prove $T_{mp} \equiv T_m \pmod{p^2}$ for $p \geq 5$. The proof uses neither a classification of Motzkin automata nor a preexisting supercongruence theorem. A related root-of-unity identity is central to the stronger results of Pan and Sun [7].

5.1 A binomial coefficient congruence

Lemma 5.1 (Babbage congruence). *For every prime p , integer $m \geq 0$, and $0 \leq j \leq m$,*

$$\binom{mp}{jp} \equiv \binom{m}{j} \pmod{p^2}. \quad (5.1)$$

Also $p \mid \binom{mp}{j}$ whenever $0 \leq j \leq mp$ and $p \nmid j$.

Proof. There is an integer polynomial $E(x)$ supported on exponents $1, \dots, p-1$ such that

$$(1+x)^p = 1 + x^p + pE(x).$$

For $m \geq 1$, expansion to first order in p gives

$$(1+x)^{mp} \equiv (1+x^p)^m + mpE(x)(1+x^p)^{m-1} \pmod{p^2}.$$

The second term has no exponent divisible by p . Extracting $[x^{jp}]$ proves (5.1). The case $m = 0$ is immediate. For the final assertion use $j \binom{mp}{j} = mp \binom{mp-1}{j-1}$ and invert j modulo p . $\square$

5.2 Why the second power of the prime appears

Let

$$\alpha = \frac{1 + \sqrt{-3}}{2}, \quad \beta = \frac{1 - \sqrt{-3}}{2}.$$

Then $\alpha + \beta = 1$, $\alpha\beta = 1$, and $\alpha^6 = \beta^6 = 1$. Moreover

$$F(x) = (x + \alpha)(x + \beta). \quad (5.2)$$

Extracting the central coefficient of the n th power gives

$$T_n = \sum_{j=0}^n \binom{n}{j}^2 \alpha^{n-j} \beta^j. \quad (5.3)$$

The two binomial coefficients, arising from the two linear factors, explain the extra p -divisibility.

To be precise about the ambient ring, work in $R = \mathbb{Z}[\alpha]$, with $\beta = 1 - \alpha$. It is a free abelian group with basis $1, \alpha$. In particular

$$p^a R \cap \mathbb{Z} = p^a \mathbb{Z}. \quad (5.4)$$

Indeed, the coefficient of α in an integer is zero, and its coefficient of 1 is that integer. Thus a congruence between integers proved coefficientwise in this basis is an ordinary integer congruence.

Theorem 5.2. *For all primes $p \geq 5$ and integers $m \geq 0$,*

$$T_{mp} \equiv T_m \pmod{p^2}. \quad (5.5)$$

In particular $T_p \equiv 1 \pmod{p^2}$.

Proof. In (5.3) with $n = mp$, any term for which $p \nmid j$ vanishes modulo p^2 , by theorem 5.1. For $j = ip$, the same lemma allows replacement of $\binom{mp}{ip}^2$ by $\binom{m}{i}^2$. Consequently

$$T_{mp} \equiv \sum_{i=0}^m \binom{m}{i}^2 (\alpha^p)^{m-i} (\beta^p)^i \pmod{p^2 R}.$$

Every prime $p \geq 5$ is 1 or 5 modulo 6. In the first case $(\alpha^p, \beta^p) = (\alpha, \beta)$; in the second it is (β, α) . The sum is therefore T_m in either case, using $i \mapsto m - i$ in the second. Finally apply (5.4). $\square$

Remark 5.3. The exclusion of $p = 3$ has a visible algebraic cause: both α^3 and β^3 equal -1 , rather than preserving or exchanging the roots. Indeed $T_3 = 7 \not\equiv 1 \pmod{9}$. This is not a technical omission in the proof.

6 Proof of (B4), multiplier extension, and sharpness

6.1 A stronger interior congruence

Theorem 6.1. *For every prime $p \geq 5$, $k \geq 2$, and $m \geq 0$,*

$$M_{mp^k+n} \equiv T_m M_n \pmod{p^2} \quad (0 \leq n \leq p^{k-1} - 3). \quad (6.1)$$

Proof. Set $s = 2$ in theorem 3.3. Its interior term is $T_{mp} M_n$, and theorem 5.2 replaces T_{mp} by T_m modulo p^2 . $\square$

Taking $m = 1$ proves (B4), since $T_1 = 1$. This completes the proof of all four OEIS-listed targets.

6.2 The first adjacent coefficient modulo p^2

Lemma 6.2. *For $p \geq 5$ prime,*

$$U_p \equiv \frac{p}{2}(1 + 3\varepsilon_p) \equiv \begin{cases} 2p, & p \equiv 1 \pmod{6}, \\ -p, & p \equiv 5 \pmod{6} \end{cases} \pmod{p^2}. \quad (6.2)$$

More generally, for $m \geq 1$,

$$U_{mp} \equiv \frac{mp}{2}(T_m + 3\varepsilon_p T_{m-1}) \pmod{p^2}. \quad (6.3)$$

Proof. Combining (2.7) with (2.3) gives the exact identity

$$2(N+1)U_N = N(T_N + 3T_{N-1}) \quad (N \geq 1). \quad (6.4)$$

For $N = mp$, the factor $N+1$ is invertible modulo p^2 , and $N/(N+1) \equiv N \pmod{p^2}$. It is enough to know the coefficients inside the parentheses modulo p .

Theorem 5.2 gives $T_{mp} \equiv T_m \pmod{p}$. Also, by Frobenius,

$$T_{mp-1} = \text{CT } P^{(m-1)p} P^{p-1} \equiv \text{CT } P(x^p)^{m-1} P(x)^{p-1} = T_{m-1} T_{p-1} \pmod{p}.$$

The last equality holds because the support of P^{p-1} is strictly between $-p$ and p . Equation (4.3) gives $T_{p-1} \equiv \varepsilon_p$. Substituting in (6.4) proves (6.3). Its case $m = 1$ is (6.2). $\square$

6.3 The maximal initial interval

Theorem 6.3 (Exact first failure modulo p^2). *For every prime $p \geq 5$ and integer $k \geq 2$,*

$$M_{p^k+p^{k-1}-2} - M_{p^{k-1}-2} \equiv \begin{cases} -2p, & p \equiv 1 \pmod{6}, \\ p, & p \equiv 5 \pmod{6} \end{cases} \pmod{p^2}. \quad (6.5)$$

In particular $p^{k-1} - 3$ is the largest upper endpoint L for which $M_{p^k+n} \equiv M_n \pmod{p^2}$ holds for every $0 \leq n \leq L$.

Proof. Use the second case of theorem 3.3 with $s = 2$, $m = 1$, and $q = p^{k-1}$. It gives

$$M_{p^k+q-2} - M_{q-2} \equiv (T_p - 1)M_{q-2} - U_p \equiv -U_p \pmod{p^2}.$$

Now apply theorem 6.2. Both resulting residues are nonzero modulo p^2 because $p \geq 5$. The preceding interior interval is valid by theorem 6.1, so its endpoint is maximal. $\square$

The assertion is about the maximal interval *starting at zero*. Some later individual offsets may again satisfy the congruence; that does not contradict the exact first failure.

Example 6.4. For $p = 5$, $k = 2$, (B4) applies to offsets 0, 1, 2. At the next offset,

$$M_{28} = 208023278209, \quad M_3 = 4,$$

so $M_{28} - M_3 \equiv 5 \pmod{25}$. For $p = 7$, $k = 2$, the first excluded offset is 5, and the difference is 35 modulo 49, equivalently -14 modulo 49. The same defect 35 occurs at offset 47 for the shift 7^3 .

7 Higher precision: the correct extension and its limitations

7.1 A transfer formula, not unrestricted periodicity

The interior of theorem 3.3 already proves

$$M_{mp^k+n} \equiv T_{mp^{s-1}} M_n \pmod{p^s}, \quad \begin{array}{l} p \text{ odd prime, } 1 \leq s \leq k, \quad m \geq 0, \\ 0 \leq n \leq p^{k-s+1} - 3. \end{array} \quad (7.1)$$

At $s = 1$ the scalar is T_m , and at $s = 2$ it can again be replaced by T_m for $p \geq 5$. At higher precision it is generally incorrect to replace it by T_m , even when $m = 1$.

For instance, choose $p = 5$, $s = k = 3$, and $n = 0$. The theorem says

$$M_{125} \equiv T_{25} \pmod{125}.$$

Exact integer computation gives

$$M_{125} \equiv T_{25} \equiv 51 \not\equiv 1 = M_0 \pmod{125}. \quad (7.2)$$

Thus even the reduced range $0 \leq n \leq p^{k-2} - 3$ does not make the unweighted analogue of (B4) true modulo p^3 . This is a counterexample to a *possible strengthening*, not to any of the four original statements.

The prime restriction in (B4) is also essential:

$$M_9 = 835 \equiv 7 \not\equiv 1 = M_0 \pmod{9}. \quad (7.3)$$

The transfer theorem correctly predicts the factor $T_3 = 7$ here.

7.2 Relationship to stronger known arithmetic

Pan–Sun’s theorem gives, for $p > 3$ and $a \geq 1$,

$$T_{mp^a} \equiv T_{mp^{a-1}} \pmod{p^{2a+2v_p(m)}} \quad (m \geq 1),$$

as a consequence of the p -adic integrality of $(T_{pN} - T_N)/(pN)^2$ [7]. This is much stronger than theorem 5.2. It can be inserted into (7.1) to reduce its scalar further. For example, for $s \geq 2$, put $t = \lfloor (s-1)/2 \rfloor$. All successive reductions from exponent $s-1$ down to t have modulus at least p^s , so

$$T_{mp^{s-1}} \equiv T_{mp^t} \pmod{p^s}. \quad (7.4)$$

For $m = 0$ the equality is immediate. This optional refinement depends on the cited stronger theorem, unlike (B1)–(B4) and the block theorem, whose proofs above are self-contained.

8 A digit algorithm for enormous indices

8.1 The Lucas rule for central trinomial coefficients

Proposition 8.1. *For a prime p , $a \geq 0$, and $0 \leq b < p$,*

$$T_{ap+b} \equiv T_a T_b \pmod{p}. \quad (8.1)$$

Consequently, if $N = d_0 + d_1 p + \cdots + d_\ell p^\ell$ with $0 \leq d_i < p$, then

$$T_N \equiv \prod_{i=0}^{\ell} T_{d_i} \pmod{p}. \quad (8.2)$$

Proof. Frobenius gives $T_{ap+b} \equiv \text{CT } P(x^p)^a P(x)^b \pmod{p}$. All exponents of $P(x)^b$ lie in $[-b, b] \subset (-p, p)$, so only constant terms can interact. This proves (8.1), and iteration gives (8.2). $\square$

This digit rule is also the foundation of the corresponding results in Kohen’s work [5, 6]. Its proof is included so that the algorithm’s correctness does not rely on a black box.

8.2 Evaluation and arithmetic complexity

For an odd prime p , first compute $t_j = T_j \bmod p$ for $0 \leq j < p$. Recurrence (2.7) can be used here because its divisors are strictly less than p . The inverses $1^{-1}, \dots, (p-1)^{-1}$ may be computed in $O(p)$ field operations using

$$\text{inv}(i) \equiv - \left\lfloor \frac{p}{i} \right\rfloor \text{inv}(p \bmod i) \pmod{p} \quad (2 \leq i < p).$$

This follows by dividing the identity $p = \lfloor p/i \rfloor i + (p \bmod i)$ by $i(p \bmod i)$ in $\mathbb{F}_p$.

Use the base- p digits of N , $N+1$, and $N+2$ to evaluate their three central trinomial coefficients, and then apply

$$M_N \equiv 2^{-1}(3T_N + 2T_{N+1} - T_{N+2}) \pmod{p}. \quad (8.3)$$

The table takes $O(p)$ storage and $O(p)$ field operations; each query then takes $O(1 + \log_p(N+2))$ field multiplications and integer digit extractions. These are arithmetic-operation counts. For a growing input integer N , the bit cost of repeated division to extract digits must also be included; the statement is not a unit-cost bit-complexity bound.

A compact version of the evaluation stage is:

```
def trinomial_from_digits(n, p, table):
    value = 1
    while n:
        n, digit = divmod(n, p)
        value = value * table[digit] % p
    return value

def motzkin_mod_prime(n, p, table):
    a = trinomial_from_digits(n, p, table)
    b = trinomial_from_digits(n + 1, p, table)
    c = trinomial_from_digits(n + 2, p, table)
    return (3*a + 2*b - c) * ((p + 1)//2) % p
```

The supplied module adds validation, construction of the table, exact arithmetic routines, and a command-line interface.

8.3 Prime-power block evaluation

For an index of the form $mp^k + d$ covered by theorem 3.3, computation modulo p^s needs only M_d , T_h , and possibly U_h , where $h = mp^{s-1}$. When k is large relative to s , these can be dramatically smaller than the target index. The implementation `block_mod_prime_power` uses precisely the three cases of (3.2).

This is a specialized block algorithm, not a general fast algorithm for arbitrary indices modulo arbitrary prime powers. Its local exact computations grow with d and h . Stating that restriction avoids confusing a useful reduction with a universal complexity result.

9 Reproducible checks and artifact contents

9.1 Independent finite verification

The accompanying verifier constructs $M_0, \dots, M_{20000}$ and enough T_n values by the exact integer recurrences. Each division is checked for a zero remainder before its result is used. It does not generate these values using the congruences under test.

The first 151 values are independently checked against the binomial sums (2.5) and (2.9); the first 81 Motzkin numbers are also checked by height-by-height path dynamic programming. The congruence checks cover odd primes up to 97, with parameters restricted so that the exact oracle contains the required indices. Multiplier bounds and the complete iteration limits are explicit in the script. The higher-precision block checks use $p \in \{3, 5, 7, 11, 13\}$, $s \leq 4$, and the same index cap.

Check family	Number of checks
Bala (B1)	251
Bala (B2)	144
Bala (B3)	98,471
Bala (B4)	10,476
Full prime-power block theorem	96,478
Prime-modulus digit algorithm	28,824
Three-trinomial identity	20,001
Prime-square multiplier extension	12,735
All remaining checks	9,434
Total	276,814

The exact first failure modulo p^2 was checked for 36 admissible prime–exponent pairs. A few rows of the full CSV are:

p	k	First excluded offset	Observed error	Modulus
5	2	3	5	25
5	3	23	5	25
7	2	5	35	49
7	3	47	35	49
11	2	9	11	121
13	2	11	143	169

These observations are consequences of theorem 6.3, not the basis of its proof. The programs are not a formal proof-assistant verification of the mathematical arguments.

9.2 Example commands

The code uses only the Python standard library and is written for Python 3.9 or later. The included run was executed with Python 3.13.5. From the archive’s top-level directory:

```
python code/verify.py
python code/motzkin.py exact 28
python code/motzkin.py mod 12345678901234567890 97
python code/motzkin.py block 1 3 0 5 3
```

The last command returns 51, illustrating (7.2). The command

```
python code/verify.py --max-index 20000 --prime-limit 97
```

recreates the checked JSON report and CSV data. Increasing the exact index cap consumes more memory because the verifier deliberately stores large exact integers; this is verification infrastructure rather than the space-efficient query algorithm.

For the much larger index $N = 10^{100} + 123456789$, the digit evaluator returns the following residues:

	p	5	7	11	97
$M_N \bmod p$	1	0	1	28	

These last outputs illustrate the proved algorithm. They were not independently compared with a construction of the enormous exact M_N , and are not included as independent oracle checks.

9.3 Files supplied

The archive contains the L^AT_EX source and its compiled PDF, `code/motzkin.py`, `code/verify.py`, `data/verification.json`, `data/sharp_boundaries.csv`, and `data/initial_values.csv`. The last file contains values through index 200, with sequence IDs and the $U_0 = 0$ convention made explicit. A README gives build and execution instructions. Notes record the source locations and a proposed OEIS update; nothing has been submitted to or changed in OEIS.

10 Conclusion

All four of Bala’s formulas follow from a small set of exact mechanisms. The factor $1 - x^2$ imposes the nonnegativity condition on paths. Frobenius and its elementary lifting spread the exponents far apart. In the interior of a block, only the central coefficient survives. At its last two offsets, a single adjacent coefficient survives as well. Finally, factorization over sixth roots of unity turns prime divisibility of a binomial coefficient into prime-square divisibility of its square.

The same reasoning settles the two natural questions that the original congruences leave implicit. The prime-5 mod 6 boundary formula alternates with the parity of the exponent, and both stated repetition ranges end exactly where the first nonzero support interaction occurs. The errors are explicit and never vanish in the relevant moduli: -1 modulo p , and either $-2p$ or p modulo p^2 . The higher-precision theorem retains the necessary central-trinomial factor rather than asserting an invalid stronger periodicity.

A An alternative rational proof of $T_p \equiv 1 \pmod{p^2}$

The root-of-unity proof gives the useful multiplier statement $T_{mp} \equiv T_m \pmod{p^2}$. For readers preferring a proof entirely in rational arithmetic, the special case needed for (B4) has the following independent derivation.

Let $L_0 = 2$, $L_1 = 1$, and $L_n = L_{n-1} - L_{n-2}$. The sequence is periodic with period six:

$$2, 1, -1, -2, -1, 1, 2, 1, \dots$$

In particular $L_p = 1$ for every prime $p \geq 5$. Its generating identity is

$$-\log(1 - z + z^2) = \sum_{n \geq 1} \frac{L_n}{n} z^n.$$

This can be checked by differentiation and the initial values. Expanding $-\log(1 - (z - z^2))$ and extracting coefficients gives, for $n \geq 1$,

$$L_n = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j \frac{n}{n-j} \binom{n-j}{j}. \quad (\text{A.1})$$

Take $n = p$. For $1 \leq j \leq (p-1)/2$,

$$\binom{p}{2j} \binom{2j}{j} = \frac{p}{2j} \binom{p-1}{2j-1} \binom{2j}{j} \equiv -\frac{p}{2j} \binom{2j}{j} \pmod{p^2}.$$

On the other hand, the j th summand of (A.1) is

$$\begin{aligned} (-1)^j \frac{p}{p-j} \binom{p-j}{j} &= (-1)^j \frac{p}{j} \binom{p-j-1}{j-1} \\ &\equiv -\frac{p}{j} \binom{2j-1}{j-1} = -\frac{p}{2j} \binom{2j}{j} \pmod{p^2}. \end{aligned}$$

All denominators in these congruences are prime to p . The $j = 0$ summand in both expressions is 1. Comparing with (2.5) therefore proves $T_p \equiv L_p = 1 \pmod{p^2}$. This also explains why the period-six distinction appears without requiring arithmetic in an extension ring.

References

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- [6] N. Kohen, *Density and Symmetry in the Generalized Motzkin Numbers mod p* , arXiv:2411.03681v1 (2024). <https://arxiv.org/abs/2411.03681>. See Proposition 6 for M_{p-2} , Proposition 7 for digit multiplicativity, and Proposition 9 for boundary digit patterns.
- [7] H. Pan and Z.-W. Sun, *Supercongruences for central trinomial coefficients*, *Frontiers in Combinatorics and Number Theory* **2** (2026), 55–62. <https://arxiv.org/abs/2012.05121v3>. First preprint posted 9 December 2020; final arXiv revision 13 March 2026. See Theorem 1.1 and Lemma 2.1.